GRID Q

An offline energy brain for a campus microgrid with a clinic on it. It forecasts, optimises and explains every 15 minutes on an Arduino UNO Q, and it keeps the clinic powered when its own software fails.

SIMULATED PLANT · REAL CONTROLLER

THE PROBLEM

Campuses and health centres on weak grids run solar, a battery and a diesel genset on fixed rules: *charge at noon, discharge in the evening, start the genset below 20%*. Rules can't see tomorrow's demand, can't tell that an outage at 19:00 costs more than one at 03:00, and can't weigh ₹27/kWh diesel against cutting a hostel. When the controller crashes, nobody is in charge.

WHAT GRIDQ DOES

GridQ replaces the rules with an **optimiser that prices every choice in rupees**: grid, battery, diesel, deferring an EV or pump, trimming a feeder. It plans 24 hours ahead against the grid failing, applies the next 15 minutes, and explains each decision. A **microcontroller that doesn't trust the optimiser** holds the safety layer.

HOW IT WORKS

01 Forecast

- Quantile gradient-boosted trees: a P50 and a P90 per feeder, 24 h ahead in one call.
- 17 causal features incl. the campus working-day calendar; P90 split-conformal calibrated.
- Solar: clear-sky × measured clearness, so it catches clouds nobody announced.

P90 coverage 90.0% held-out · solar error 0.26 vs 0.48 kW for persistence

02 Optimise

- A linear program on HiGHS, re-solved every 15 min as **MPC**: 24 h of battery physics, a 40 kW cap, fuel, deadlines, fair shedding.
- **Outage fan tree**: plans against the grid failing at +15 min…+4 h, weighted by real outage statistics, with non-anticipativity.
- Every "why" is a re-solve or an LP dual: *"EV paused, running it now costs ₹17 more."*

5,837 variables · 1,557 constraints · 48 ms median laptop, ~0.1 s on UNO Q

03 Fail safe

- The optimiser is a separate, killable process and the **only** source of heartbeats.
- The STM32 MCU goes SAFE (clinic + pump only) 3 s after the last *new* heartbeat.
- The MCU trips the grid relay from the grid-sense pin itself, with no Linux in the loop. The relay is wired fail-safe.

KILL → SAFE in < 3.5 s · local trip, 50 ms debounce · reclose after 5 s

RESULTS ON HELD-OUT REAL DATA

DEC 2017 · 100 DAYS	TOTAL COST	UNSERVED	CLINIC	FAIRNESS
B0 · inverter default	₹7,83,106	1,489 kWh	100%	0.38
B1 · tuned time-of-day rules	₹8,05,486	1,335 kWh	100%	0.44
GridQ	**₹7,67,968**	**317 kWh**	**100%**	**1.00**
Oracle · perfect foresight	₹7,48,038	17 kWh	100%	1.00

Real I-BLEND campus demand (IIIT-Delhi) and NASA POWER solar on days the forecaster never saw, each with 5 random outage patterns. Tuned on days 1–10, reported on 11–30. Total cost = energy + diesel + demand charge + battery wear + value of lost load. Fairness = Jain index.

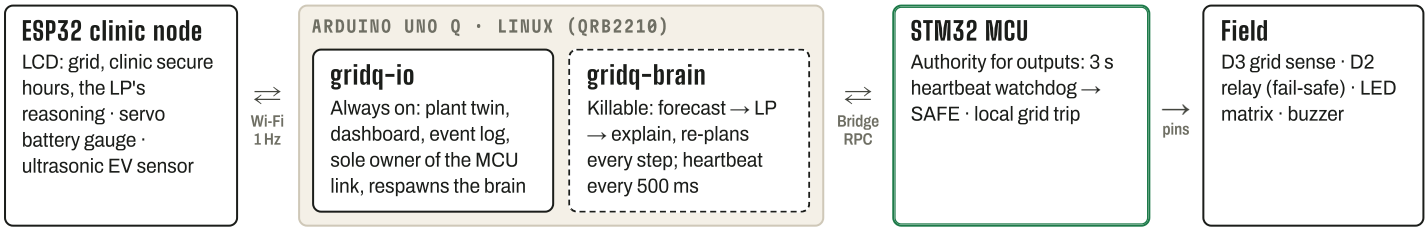
43% of the achievable saving captured vs the stronger baseline; 65% vs B1 (the spec's measure)

−79% unserved energy vs B0, shared fairly instead of always cutting one hostel

93% captured in the summer stress block, when the 40 kW grid cap binds

67.5% captured by the outage fan tree on tune days; the spec's fixed clinic reserve scored −38.6%

ARCHITECTURE · WHATEVER CAN CRASH IS NOT IN CHARGE OF SAFETY



BREAK IT, LIVE

Pull the D3 jumper
The MCU islands the site by itself: relay clicks, buzzer sounds. The LP re-plans on battery + diesel and trims by price, with the clinic last.

Press KILL
SIGKILL the optimiser. Heartbeats stop, the MCU goes SAFE in under 3.5 s, the brain respawns and NORMAL returns on its own.

Cloud · EV fleet
Solar ×0.3 unannounced, or a hand over the ultrasonic sensor adds 30 kWh due by 06:00. The plan adapts on the next step.

Battery fault
Capacity and power halve. The plan, the clinic's secure hours and the servo gauge all respond at once.

REAL VS SIMULATED

REAL — Forecaster, LP/MPC brain, explanations, MCU safety supervisor, relay, LED matrix, buzzer, ESP32 clinic node, demand and solar data.

SIM — The power plant (30 kWp solar, 50 kWh battery, 24 kW genset, feeders). Clinic demand is modelled and labelled.

WHY IT MATTERS

Runs **fully offline** on a credit-card-sized board, with no cloud and no deep nets or LLMs. Every decision has a rupee explanation an operator can check, and every number here is computed by the repo. Swap the plant twin for real meter readings and the same LP and safety logic run a real site.